MAT-MEK4270 Mandatory Assignment 1

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1.2.4 Dispersion coefficient

Assume $m_x = m_y$, so that $k_x = k_y = k$. Consider the discrete wave

$$u_{ij}^n = e^{i(kh(i+j)-\widetilde{\omega} n\Delta t)}.$$

We insert this into the central difference scheme

$$u_{ij}^{n+1} - 2u_{ij}^n + u_{ij}^{n-1} = C^2\big(u_{i+1,j}^n + u_{i-1,j}^n + u_{i,j+1}^n + u_{i,j-1}^n - 4u_{ij}^n\big),$$

where $C = c\Delta t/h$.

The time shifts satisfy

$$u_{ij}^{n+1} = e^{-i\widetilde{\omega}\Delta t}u_{ij}^n, \qquad u_{ij}^{n-1} = e^{i\widetilde{\omega}\Delta t}u_{ij}^n.$$

The spatial shifts satisfy

$$u_{i\pm1,j}^n = e^{\pm ikh}u_{ij}^n, \qquad u_{i,j\pm1}^n = e^{\pm ikh}u_{ij}^n.$$

Substituting and dividing by the nonzero u_{ij}^n gives

$$e^{-i\widetilde{\omega}\Delta t} - 2 + e^{i\widetilde{\omega}\Delta t} = 2C^2\left(e^{ikh} - 2 + e^{-ikh}\right).$$

Using

$$e^{i\theta} + e^{-i\theta} - 2 = -4\sin^2(\theta/2),$$

we obtain

$$\boxed{\sin^2\left(\frac{\widetilde{\omega}\Delta t}{2}\right) = 2C^2\sin^2\left(\frac{kh}{2}\right)}.$$

For $C = 1/\sqrt{2}$, this becomes

$$\sin^2\left(\frac{\widetilde{\omega}\Delta t}{2}\right) = \sin^2\left(\frac{kh}{2}\right).$$

For a resolved wavenumber $0 \le kh \le \pi$, choosing the nonnegative principal frequency branch gives

$$\widetilde{\omega}\Delta t = kh.$$

Therefore,

$$\widetilde{\omega} = \frac{kh}{\Delta t} = \frac{ck}{C} = \sqrt{2}\,ck.$$

The exact frequency for $k_x = k_y = k \ge 0$ is

$$\omega = c\sqrt{k_x^2 + k_y^2} = \sqrt{2}\,ck.$$

Thus, for $C = 1/\sqrt{2}$,

$$\boxed{\widetilde{\omega} = \omega}.$$